MA 442 - practice problems for exam 1

February 22

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A collection of problems designed to help you prepare for midterm exam 1. These problems are by no means exhaustive, but they are in the same style that will appear on the exam.

Question 1. Let V be a finite-dimensional vector space over $\mathbb{R}$. In class we (will) show that the set of *all* linear maps $\mathsf{V} \to \mathbb{R}$ is naturally a vector space; we denote this vector space by $\mathrm{Hom}(\mathsf{V}, \mathbb{R})$. Express

$$\dim \mathrm{Hom}(\mathsf{V}, \mathbb{R})$$

in terms of $\dim V$. (Hint: Given a basis for V can you construct a basis for $\mathrm{Hom}(\mathsf{V}, \mathbb{R})$?)

Solution. We will show that $\dim \mathrm{Hom}(\mathsf{V}, \mathbb{R}) = \dim \mathsf{V}$. Let $\dim \mathsf{V} = n$, and fix a basis $\{v_i\}$ for V. For each $i = 1, \ldots, n$ we will define a linear transformation $\varphi_i \colon \mathsf{V} \to \mathbb{R}$. Note that since we have fixed a basis, to define φ_i on all of V it suffices to define it only on the basis elements. So, define

$$\varphi_i(v_j) = \begin{cases} 1, & i = j \\ 0, & \text{else} \end{cases} \tag{1}$$

We will show that $\{\varphi_1, \ldots, \varphi_n\}$ form a basis for $\mathrm{Hom}(\mathsf{V}, \mathbb{R})$. First, we show that this set generates. Suppose $\varphi \colon \mathsf{V} \to \mathbf{R}$ is an arbitrary linear transformation. Since it is a linear transformation, it is completely determined by its behavior on a basis. Thus, φ is completely determined by its values on the vectors $\{v_1, \ldots, v_n\}$. Call these values $a_i \overset{\text{def}}{=} \varphi(v_i)$ for $i = 1, \ldots, n$. Then, we observe that

$$\varphi = \sum_i a_i \varphi_i. \tag{2}$$

Indeed, if we evaluate the left hand side at v_j then this is a_j by definition. On the other hand, since $\varphi_i(v_j)$ is 1 when $i = j$ and 0 otherwise, the right hand side agrees. Thus, we have shown that the collection $\{\varphi_i\}$ generates $\mathrm{Hom}(\mathsf{V}, \mathbb{R})$.

It remains to show that this collection of vectors is linearly independent. Suppose we have a linear relation

$$\lambda_1 \varphi_1 + \cdots + \lambda_n \varphi_n = \mathbf{0} \tag{3}$$

where the right hand side is the zero linear transformation. If we evaluate this as v_1 then the left hand side is λ_1. Thus $\lambda_1 = 0$. Similarly, we can evaluate at $v_2, v_3, \ldots, v_n$ to see that $\lambda_2 = \lambda_3 = \cdots = \lambda_n = 0$ as well. This shows that $\{\varphi_i\}$ is indeed a basis.

Question 2. Let V be any vector space and suppose that $B = \{v_1, v_2, v_3, v_4\}$ is an ordered basis. Define the linear transformation $\mathsf{T} \colon \mathsf{V} \to \mathsf{V}$ by the formulas

$$\mathsf{T}(v_1) = v_1 + v_2, \ \mathsf{T}(v_2) = v_2 + v_3, \ \mathsf{T}(v_3) = v_3 + v_4, \ \mathsf{T}(v_4) = v_4.$$

Find the matrix representation of T with respect to this ordered basis. (So, compute $[\mathsf{T}]_B$.)

Solution.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

Question 3. Suppose that $\mathsf{T}\colon \mathbb{R}^2 \to \mathbb{R}^2$ linear transformation with the property that $\dim\ker \mathsf{T} = 1$. Prove that there exists two bases β and γ of $\mathbb{R}^2$ such that the matrix representation of T is

$$[\mathsf{T}]_\beta^\gamma = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}. \tag{4}$$

Solution. By assumption, there exists a nonzero $v \neq 0$ such that $\mathsf{T}(v) = 0$. In particular $\{v\}$ is a linearly independent subset of $\mathbb{R}^2$. We can extend to a basis $\{v, u\}$ of $\mathbb{R}^2$, which we will reorder to view as the ordered basis $\beta = \{u, v\}$. Notice that $\mathsf{T}(u) \neq 0$ since $\dim\ker \mathsf{T} = 1$ and u, v are linearly independent by construction. Thus $\{\mathsf{T}(u)\}$ is a linearly independent subset of $\mathbb{R}^2$. We can, in turn, extend it to an ordered basis $\gamma = \{\mathsf{T}(u), w\}$ of $\mathbb{R}^2$. These choices of bases result in the desired matrix.

Question 4. Recall that $\mathbb{R}[x]$ is the vector space of all real polynomials. Let $\mathsf{E} \subset \mathbb{R}[x]$ be the subset of *even* polynomials (so, only even powers of x can appear, like $3 + 4x^6 + 9x^{38}$). Let $\mathsf{U} \subset \mathbb{R}[x]$ be the subset of *odd* polynomials (so, only odd powers of x can appear like $2x + x^7 + 6x^{29}$.)

(a) Prove that both E and U are subspaces of $\mathbb{R}[x]$.

(b) Prove that $\mathbb{R}[x] = \mathsf{E} \oplus \mathsf{U}$.

Question 5. Prove that there exists a linear transformation $\mathsf{T}\colon \mathbb{R}^2 \to \mathbb{R}^3$ such that $\mathsf{T}(1,1) = (1,0,2)$, $\mathsf{T}(2,3) = (1,1,3)$. Also, compute $\mathsf{T}(7,10)$.

Question 6. This question has two parts.

(a) Suppose that W_1 and W_2 are two two-dimensional subspaces in $\mathbb{R}^3$. Prove that their intersection must contain a nonzero vector. In other words, prove that $\mathsf{W}_1 \cap \mathsf{W}_2 \neq \{0\}$.

(b) Provide an example of two two-dimensional subspaces in $\mathbb{R}^4$ which intersect at single point.

In other words, two planes cannot intersect at a single point in $\mathbb{R}^3$ but they can in $\mathbb{R}^4$!

Question 7. Let $\mathbb{R}[x]$ denote the vector space of real-valued polynomials. Consider the linear maps

$$\partial, \partial^2, \partial^3 \colon \mathbb{R}[x] \to \mathbb{R}[x] \tag{5}$$

defined by the formulas $\partial(f) = f'$, $\partial^2(f) = f''$, $\partial^3(f) = f'''$. Show that the four-element subset

$$\{\mathbb{1}, \partial, \partial^2, \partial^3\} \subset \mathrm{Hom}(\mathbb{R}[x], \mathbb{R}[x])$$

is linearly independent.

Question 8. Suppose that V is a real vector space (a vector space defined over the scalars $\mathbb{R}$). Also, suppose that $\mathsf{W} \subset \mathsf{V}$ is a subspace with a finite number of elements. Show that $\mathsf{W} = \{0\}$.

Question 9. Let P_3 be the vector space of real-valued polynomials which are at most cubic. Let $\mathsf{W} \subset \mathsf{P}_3$ be the subset of polynomials f such that $f(2) = 0$.

(a) Show that W is a subspace.

(b) Find a basis for W.

Question 10. Can you find a linear transformation $\mathsf{T}\colon \mathbb{R}^2 \to \mathbb{R}^2$ such that $\operatorname{Im}\mathsf{T} = \ker\mathsf{T}$? If yes, construct an explicit example. If no, prove that there cannot exist one.

Question 11. The collection of vectors

$$\left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \end{bmatrix} \right\}.$$

is a basis for $\mathbb{R}^2$. Express the standard unit vectors $\{e_1, e_2\}$ in terms of these basis vectors.